

# SERAT M. SAAD

Ph.D. Student in Astronomy, The Ohio State University

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## RESEARCH INTERESTS

**Dynamics:** binary stars, radial migration, tests of Modified Newtonian Dynamics (MOND), an alternative to dark matter

**Stellar Spectroscopy:** chemical abundances, metal-rich stars, star spots, young stellar objects

**AI for Astronomy:** LLM agents, agent coordination, automation of astrophysical research

## EDUCATION

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Ph.D. in Astrophysics (in progress), The Ohio State University<br><i>Advisors: Prof. Yuan-Sen Ting and Prof. Krzysztof (Kris) Stanek</i>                                                         | 2025-Present |
| A.B. in Physics & Applied Mathematics, Vanderbilt University<br><i>Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra</i><br><i>Advisor: Prof. Keivan Stassun</i> | 2021-2025    |

## HONORS & AWARDS

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|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| University Distinguished Fellowship, The Ohio State University<br><i>graduate-school fellowship supporting doctoral study</i>                              | 2029-2030 |
| University Fellowship, The Ohio State University<br><i>graduate-school fellowship supporting doctoral study</i>                                            | 2025-2027 |
| Hult Prize, National Finalist<br><i>for a student venture expanding access to personalized education</i>                                                   | 2026      |
| Larry R. Cathey Award, Vanderbilt University<br><i>annual departmental award for the top graduating student studying astronomy</i>                         | 2025      |
| Highest Honors in Astronomy, Vanderbilt University<br><i>for distinction in the undergraduate honors thesis and its defense</i>                            | 2025      |
| International Full-ride Student Grant, Vanderbilt University<br><i>full four-year scholarship covering all costs of undergraduate study</i>                | 2021-2025 |
| Dean's List, Vanderbilt University<br><i>eight semesters of academic distinction</i>                                                                       | 2021-2025 |
| Buchanan Fellowship, Vanderbilt University<br><i>supported a data-science project applying NLP to British periodical collections</i>                       | 2022      |
| Silver Medal, International Olympiad on Astronomy and Astrophysics<br><i>represented Bangladesh at IOAA 2019 and the Global e-Competition (GeCAA) 2020</i> | 2020      |

## PUBLICATION SUMMARY

13 papers, 7 as first author, 7 published or accepted in refereed journals.

108 citations from 104 citing papers ([NASA ADS Library](#)).

## TELESCOPE TIME ALLOCATIONS

### Principal Investigator

|                                                                              |       |
|------------------------------------------------------------------------------|-------|
| Large Binocular Telescope, PEPSI, 20 hours · PANTERA: ultra metal-rich stars | 2026B |
| Large Binocular Telescope, PEPSI, 20 hours · PASTA: 36 planet-hosting stars  | 2026A |
| Large Binocular Telescope, PEPSI, 28 hours · PASTA: 43 planet-hosting stars  | 2025B |

### Co-Investigator

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|-----------------------------------------------------------------------------------------|-------|
| WIYN, NEID, NN-EXPLORE Key Program · 65 HWO target stars (PI: M. C. Johnson, submitted) | 2026  |
| Automated Planet Finder, Levy, 40 hours · metal-rich stars (PI: D. M. Rowan)            | 2026B |
| Chandra X-ray Observatory, Cycle 26 · Pink Mergers (PI: Prof. Adi Foord)                | 2024  |

## INVITED TALKS AND CONFERENCE PRESENTATIONS

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|-------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Monday Science Seminar, University of Wisconsin–Madison<br><i>Title to be announced</i>                                                               | 2026 |
| LLM Agents Seminar, <a href="#">NASA AI/ML STIG</a><br><i>Hands-on session II: agentic coding and research tools · recording · slides</i>             | 2026 |
| AAS 245 Winter Meeting, American Astronomical Society<br><i>An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)</i> | 2025 |
| Cool Stars 22, Cambridge<br><i>Using Low Resolution Optical Spectra to Identify Young Stars and Their Properties</i>                                  | 2024 |
| IEEE Quantum Computing and Engineering (QCE)<br><i>Developing Workforce in Quantum Industry: The Wond’ry Quantum Studio</i>                           | 2023 |

## TEACHING & MENTORING

*Teaching and grading in astronomy, statistics, data science, and programming, alongside mentoring olympiad students in Bangladesh and graduate-school applicants from underrepresented backgrounds.*

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Mentor, <a href="#">Project SHORT</a><br><i>free admissions mentoring for underrepresented applicants to graduate and medical programs</i>                                               | 2025        |
| Mentor, <a href="#">Bangladesh Olympiad on Astronomy and Astrophysics</a><br><i>national competition that selects and trains Bangladesh’s team for the international olympiad (IOAA)</i> | 2021-2024   |
| TA, DS 2100: Statistics for Data Science, Vanderbilt University                                                                                                                          | Spring 2025 |
| TA, ASTR 1010: Introductory Astronomy, Vanderbilt University                                                                                                                             | Spring 2024 |
| Grader, DS 1100: Introduction to Data Science, Vanderbilt University                                                                                                                     | Fall 2023   |
| Instructor, Quantum Computing for Everyone, <a href="#">The Wond’ry</a> Vanderbilt                                                                                                       | Fall 2023   |
| Grader, CS 1100: Introduction to Problem Solving with Python, Vanderbilt University                                                                                                      | 2022-2023   |
| Tutor, Vanderbilt Student Athletics                                                                                                                                                      | 2022        |

## OTHER EXPERIENCES

*Building an educational AI initiative, communicating new research through writing, and several years training Bangladesh’s astronomy olympiad team.*

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Co-founder, <a href="#">PorAI</a><br><i>educational AI initiative bringing personalized, tutor-like instruction to students and schools</i>                                              | 2026-Present |
| Author and Editor, <a href="#">Astrobites</a>                                                                                                                                            | 2026-Present |
| Journal Referee (A&A)                                                                                                                                                                    | 2026-        |
| Observer, Large Binocular Telescope, 5 nights                                                                                                                                            | 2026         |
| Academic Team Member, <a href="#">Bangladesh Olympiad on Astronomy and Astrophysics</a>                                                                                                  | 2021-2024    |
| Co-lead, <a href="#">The Wond’ry</a> Quantum Studio<br><i>student-led quantum computing program at Vanderbilt’s innovation center</i>                                                    | 2022-2023    |
| Team Leader, <a href="#">International Olympiad on Astronomy and Astrophysics</a><br><i>led Bangladesh’s national team and served as a question moderator at the international round</i> | 2022         |

## FIRST AUTHOR PUBLICATIONS AND PREPRINTS

7. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“More than Half of Recent Astronomy Papers are Written with Language-Model Assistance”  
[arXiv:2609.10664](#)
6. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra”  
[The Open Journal of Astrophysics, submitted](#)
5. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek, et al. (2026)  
“PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy”  
[The Open Journal of Astrophysics, submitted](#)
4. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“No Gravitational Anomaly in Wide Binaries from Forward Modeling of 3D Orbits”  
[The Open Journal of Astrophysics, Vol. 9](#)

3. **Serat M. Saad** & Yuan-Sen Ting (2026)  
“High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND)”  
*The Open Journal of Astrophysics, Vol. 9*
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun, et al. (2025)  
“ABYSS III: Observing Accretion Activity in Young Stars through Empirical Veiling Measurements”  
*The Astronomical Journal, Vol. 170, 87*
1. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun, et al. (2024)  
“ABYSS II: Identification of Young Stars in Optical SDSS Spectra and Their Properties”  
*The Astronomical Journal, Vol. 167, 125*

## CO-AUTHOR PUBLICATIONS AND PREPRINTS

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6. Yuan-Sen Ting & **Serat M. Saad** (2026)  
“Armillary: A Label-Free Coordinate System for Stellar Spectra”  
*The Open Journal of Astrophysics, submitted*
5. Qinghui Sun, **Serat M. Saad**, Yuan-Sen Ting, et al. (2026)  
“Planets Around Solar Twins/Analog (PASTA) III: Chemical Clock Relations in Planet-Hosting Solar Analogs”  
*Astronomy & Astrophysics, accepted*
4. SDSS Collaboration, including **Serat M. Saad** (2026)  
“The Twentieth Data Release of the Sloan Digital Sky Surveys”  
*The Astrophysical Journal Supplement Series, submitted*
3. Zimo Cheng, Sharon X. Wang, Fan Liu, . . . **Serat M. Saad**, et al. (2026)  
“A Bayesian Search for Planet Engulfment Signatures in Solar Analogs”  
*AAS Journals, submitted*
2. SDSS Collaboration, including **Serat M. Saad** (2026)  
“The Nineteenth Data Release of the Sloan Digital Sky Survey”  
*The Astrophysical Journal Supplement Series, Vol. 285, 9*
1. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu, Yuting Shen (2026)  
“Egent: An Autonomous Agent for Equivalent Width Measurement”  
*The Open Journal of Astrophysics, Vol. 9*

## NON-REFEREED PUBLICATIONS

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6. **Serat M. Saad** (2026)  
“‘Total Eclipse of the Heart’ in Astronomy”  
*Astrobites, August*
5. **Serat M. Saad** (2026)  
“Where Have All the Star Couples Gone?”  
*Astrobites, August*
4. **Serat M. Saad** (2026)  
“Cosmic Cannibalism: When Stars Eat Their Planets”  
*Astrobites, May*
3. **Serat M. Saad** (2026)  
“Wormholes Might Be More Real Than We Thought”  
*Astrobites, April*
2. **Serat M. Saad** (2026)  
“A Wandering Supermassive Black Hole Eating a Star”  
*Astrobites, March*
1. **Serat M. Saad**  
“Pixels and Persuasion”  
*As Seen on TV, digital exhibition, Vanderbilt University Libraries*

## REFERENCES

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|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Krzysztof (Kris) Stanek, The Ohio State University<br><i>University Distinguished Scholar; Guggenheim Fellow (2022)</i>                    | Ph.D. Advisor, <a href="mailto:stanek.32@osu.edu">stanek.32@osu.edu</a>                                 |
| Yuan-Sen Ting, The Ohio State University<br><i>Inaugural Chair, NASA AI/ML Science and Technology Interest Group; Kavli Scholar (2026)</i> | Ph.D. Advisor, <a href="mailto:ting.74@osu.edu">ting.74@osu.edu</a>                                     |
| Keivan G. Stassun, Vanderbilt University<br><i>National Medal of Science (2024); MacArthur Fellow (2024)</i>                               | Undergraduate Advisor, <a href="mailto:keivan.stassun@vanderbilt.edu">keivan.stassun@vanderbilt.edu</a> |